

PROFESSIONAL SYLLABUS

Fundamentals for Beginners

A beginner-friendly course covering essential networking and cybersecurity concepts — how networks work, common threats, protective technologies, and hands-on traffic analysis using industry-standard tools in a safe lab environment.

MODULES

10 Modules

LEVEL

**Beginner
Friendly**

FORMAT

**Theory + Hands-
On**

OUTCOME

SOC-Ready Base



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Course Description

Network security is the practice of protecting computers, devices, and data while they communicate across networks such as the internet or an organization's internal network. This beginner-friendly course introduces the **essential concepts of networking and cybersecurity**, helping students understand how networks work, common security threats, and the techniques used to protect digital systems.

The course combines **theory with hands-on activities**, enabling learners to gain practical experience using industry-standard security tools in a safe lab environment.

Course Objectives

- ✓ Understand basic networking concepts and terminology.
- ✓ Identify common network security threats and attacks.
- ✓ Explain how firewalls, antivirus software, and encryption protect systems.
- ✓ Analyze basic network traffic using security tools.
- ✓ Apply security best practices for home and business networks.
- ✓ Build a foundational understanding for advanced cybersecurity studies.

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Course Modules

MODULE 01 Introduction to Networking & Cybersecurity

TOPICS

- What is a Network?
- Types of Networks (LAN, WAN, Internet)
- What is Cybersecurity?
- Why Network Security Matters
- Real-World Cyber Incidents

LEARNING OUTCOMES

- Understand basic networking concepts.
- Explain why networks need protection.
- Recognize common cyber risks.

HANDS-ON ACTIVITY Draw a simple home or college network diagram.

MODULE 02 How Data Travels Across Networks

TOPICS

- IP Addresses
- MAC Addresses
- DNS Basics
- Routers and Switches
- TCP/IP Fundamentals

LEARNING OUTCOMES

- Understand how devices communicate.
- Identify common networking components.
- Explain the journey of data packets.

HANDS-ON ACTIVITY Use command-line tools to view IP addresses and network information.

MODULE 03 Network Devices & Security Components**TOPICS**

- Routers
- Switches
- Firewalls
- Wireless Access Points
- Security Appliances

LEARNING OUTCOMES

- Identify network devices.
- Understand their security functions.
- Explain traffic control concepts.

HANDS-ON ACTIVITY Create a network topology diagram showing security devices.

MODULE 04 Common Network Threats & Attacks**TOPICS**

- Malware
- Viruses and Worms
- Phishing
- Denial of Service (DoS)
- Password Attacks

LEARNING OUTCOMES

- Recognize common cyber threats.
- Understand attack motivations.
- Identify warning signs of attacks.

HANDS-ON ACTIVITY Analyze sample phishing emails and identify red flags.

MODULE 05 Firewalls & Access Control**TOPICS**

- What is a Firewall?
- Network Filtering Concepts
- Access Control Basics
- Security Policies
- Safe Internet Usage

LEARNING OUTCOMES

- Explain firewall functionality.
- Understand traffic filtering.
- Describe basic access control methods.

HANDS-ON ACTIVITY Create simple firewall rule examples using scenarios.

MODULE 06 Encryption & Secure Communication**TOPICS**

- What is Encryption?
- HTTPS
- VPN Basics
- Digital Certificates
- Password Security

LEARNING OUTCOMES

- Understand how encryption protects data.
- Explain secure communication concepts.
- Identify encrypted services.

HANDS-ON ACTIVITY Explore HTTPS-secured websites and certificate information.

MODULE 07**Network Monitoring & Traffic Analysis****TOPICS**

- Network Monitoring Concepts
- Packet Analysis
- Logging Basics
- Security Alerts
- Incident Detection

LEARNING OUTCOMES

- Understand network visibility.
- Interpret basic traffic information.
- Recognize suspicious activity.

HANDS-ON ACTIVITY Capture and inspect network traffic using Wireshark.

MODULE 08**Wireless Network Security****TOPICS**

- Wi-Fi Fundamentals
- Wireless Risks
- Secure Wi-Fi Configuration
- WPA2 / WPA3 Security
- Public Wi-Fi Safety

LEARNING OUTCOMES

- Understand wireless security risks.
- Configure secure wireless settings.
- Apply safe wireless practices.

HANDS-ON ACTIVITY Perform a wireless security checklist assessment.

MODULE 09**Vulnerability Awareness & Security Best Practices****TOPICS**

- Software Updates
- Vulnerabilities Explained
- Security Misconfigurations
- Password Management
- Multi-Factor Authentication

LEARNING OUTCOMES

- Identify common weaknesses.
- Apply basic security practices.
- Improve system security posture.

HANDS-ON ACTIVITY Conduct a security assessment of a sample workstation.

MODULE 10**Introduction to Security Operations & Career Paths****TOPICS**

- Security Monitoring
- Incident Response Basics
- Security Team Roles
- Security Policies
- Cybersecurity Career Roadmap

LEARNING OUTCOMES

- Understand how security teams operate.
- Learn basic incident response concepts.
- Explore cybersecurity career opportunities.

HANDS-ON ACTIVITY Create a simple incident response plan for a small business.

Practical Components & Tools

Students gain hands-on exposure to beginner-friendly, industry-standard security tools used in real-world network defense:

- **Wireshark**

Network traffic capture & deep packet analysis.

- **Nmap**

Basic network scanning & service discovery demos.

- **VirtualBox**

Creating safe, isolated lab environments for practice.

- **Cisco Packet Tracer**

Visual network simulations & topology design.

- **OS Network Utilities**

Ping, Traceroute, IP Configuration tools across Windows & Linux.

- **Browser Security Tools**

Inspecting HTTPS, certificates & secure connections.

MINI PROJECT

Design and Secure a Small Office Network

- ✓ Create a complete network diagram
- ✓ Identify security risks & weak points
- ✓ Recommend firewall rules & access controls
- ✓ Implement security best practices for users & devices

Assessment & Grading

Final evaluation is based on a balanced mix of quizzes, lab work, a case study, and a capstone mini project.

COMPONENT	WEIGHTAGE
Module Quizzes Short quizzes covering key concepts.	30%
Practical Activities Completion of hands-on exercises and lab tasks.	30%
Network Security Case Study Analysis of a real-world cyber incident.	20%
Final Mini Project Design and present a secure network solution for a small business.	20%
Total	100%

Recommended Resources

Curated books, free learning platforms, and reference sites to deepen your network security knowledge.

Books

- Network Security Essentials — William Stallings
- CompTIA Security+ Study Guide (Beginner Sections)

Free Learning Platforms

- Cisco Networking Academy — netacad.com
- TryHackMe — Beginner Learning Paths
- Cybrary — Free Courses

📖 Reference Websites

- CISA Cybersecurity Resources — cisa.gov
- OWASP Foundation — owasp.org

🔧 Practice Labs

- VirtualBox / VMware Workstation Player
- Kali Linux (training distribution)
- Wireshark sample capture files

Course Outcome & Next Steps

Upon completion, learners will understand how computer networks function, recognize common cyber threats, analyze basic network traffic, and apply **foundational security measures**. This course serves as an excellent stepping stone toward **Network Administration, Ethical Hacking, Cybersecurity Analysis, and Security Operations Center (SOC)** training.

Recommended next learning tracks at Obscyratech:

[Ethical Hacking](#)[VAPT](#)[SOC Analyst Training](#)[Cybersecurity Analysis](#)[Network Administration](#)[CompTIA Security+](#)

Program Promise: This Network Security Fundamentals course is designed for absolute beginners — including students, IT enthusiasts, and career switchers — to build a **solid networking and security foundation** that prepares them for advanced cybersecurity learning paths and entry-level industry roles.

OBSCYRATECH

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Master the basics of — the foundation of every cyber career.

Join Obscyratech's Network Security Fundamentals program and gain the hands-on knowledge needed to protect networks, analyze traffic, and step into advanced cybersecurity paths like Ethical Hacking, VAPT, and SOC operations.

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LOCATION

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PROGRAM

**Network Security
Fundamentals**